Ensor Palacios, Ph.D.

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Education

2018-2023 **University of Bristol**
PhD Wellcome Trust Neural Dynamics Programme.
Project and supervisors: excitatory-inhibitory balance and information processing in the cerebellar cortex; Conor Houghton, Paul Chadderton.

2018 **University College London**
Erasmus internship.
Project and supervisor: simulations of neural network self-organisation under the free energy principle; Karl Friston.

2015-2017 **University of Padua**
Master in Cognitive Neuroscience and Clinical Neuropsychology.

Project and supervisors: simulations of biological self-organisation under the free energy principle; Karl Friston, Gianluca Campana.

2012-2015 **University of Padua**
Bachelor's degree in Cognitive Psychology and Psychobiology.

Employment

2024-present **University of Bath** – Department of Mathematical Sciences.
Research associate in the group of Dr. Theresa Smith working on health-related problems using statistical and machine learning techniques; position funded by the AI4CI (UKRI AI Research hub).

2024 **University of Bristol** – MRC Integrative Epidemiology Unit.
Postdoctoral transition fellowship in epidemiology, investigating the effects of air pollution on dementia in the UK Biobank cohort; project lead in collaboration with Prof. Kate Tilling.

Funds and awards

2026 Impact fund, University of Bath (£2500).

2026 Seedcorn fund, Cabot Institute for the Environment (£5000).

2024 Stipend, Wellcome Trust funded transition fellowship (8 months).

2018 Stipend, Erasmus Programme (9 months).

2017 Stipend, Erasmus Programme (7 months).

Research Skills and Experience

R, Python, Shiny

Time series

Survival analysis

Causal inference

Github

https://github.com/ensorpalacios

Publications

2025 **Palacios E. R.**, Shapland C. Y., Wolf L. J., Nordestgaard L. T., Anderson E., Slaney C., Bernie D., Mitchell D., Kehoe P. G., Griffith G. J., Tilling K., Dissecting the effect of long-term exposure to air pollution on risk of dementia in UK Biobank, *Environmental Health*, 2025. https://doi.org/10.1186/s12940-025-01096-3

Palacios E. R., Houghton C., Chadderton P., GlyT2-Positive Interneurons Regulate Timing and Variability of Information Transfer in a Cerebellar-Behavioral Loop, *Journal of Neuroscience*, 2025. https://doi.org/10.1523/JNEUROSCI.0000-24.2025

2024 C. C. Slaney, N. M. Giollabhui, P. J. Van der Most, E. R. Palacios, H. Snieder, M. Nivard, G. Hemani, C. A. Hartman, G. M Khandaker, Positive and negative affect, related mental health traits, and cognitive performance: shared genetic architecture and potential causality, *medRxiv*, 2024. https://doi.org/10.1101/2024.XX.XXXXXX

Palacios E. R., Chadderton P., Friston K., Houghton C., Cerebellar state estimation enables resilient coupling across behavioural domains, *Scientific Reports*, 2024. https://www.nature.com/articles/s41598-024-56811-x

2022 **Palacios E. R.**, Houghton C., Chadderton P., The role of Golgi cells in cerebellar cortical transformation, *SFN Poster*, 2022. https://zenodo.org/record/7304821

2021 **Palacios E. R.**, Houghton C*., Chadderton P.*, Accounting for uncertainty: inhibition for neural inference in the cerebellum, *Proceedings of the Royal Society B*,

2021. https://doi.org/10.1098/rspb.2021.0276

2020 **Palacios E. R.**, Razi A., Parr T., Kirchhoff M., Friston K., On Markov blankets and hierarchical self-organisation, *Journal of Theoretical Biology*, 2020. https://doi.org/10.1016/j.jtbi.2019.110089

2019 **Palacios E. R.**, Isomura T., Parr T., Friston K., The emergence of synchrony in networks of mutually inferring neurons, *Scientific Reports*, 2019. https://doi.org/10.1038/s41598-019-42821-7

2018 Kirchhoff M., Parr T., **Palacios E. R.**, Friston K., Kiverstein J., The Markov blankets of life: autonomy, active inference and the free energy principle, *Journal of the Royal Society Interface*, 2018. https://doi.org/10.1098/rsif.2017.0792

**Equal contribution.*

Conferences and talks

2025 ISCB, poster presentation.

2023 Najafi Lab (Georgia Tech), invited talk.

2022 SFN, poster presentation.